**Problem 1 part 1)**

Given the image region as shown in Figure 1 and $V = \{1\}$, what is the shortest m-path between $p$ (the pixel at the upper-left corner) and $q$ (the pixel at the bottom-right corner)? Give the length of the path if the path exists. (15 points)

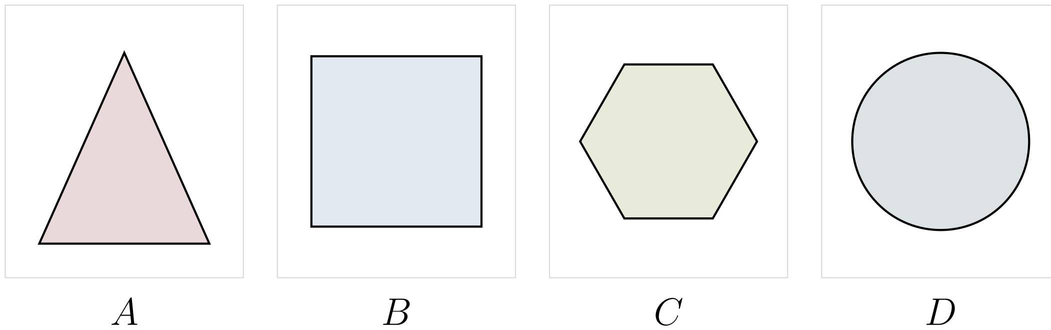
$$\begin{matrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 \end{matrix}$$

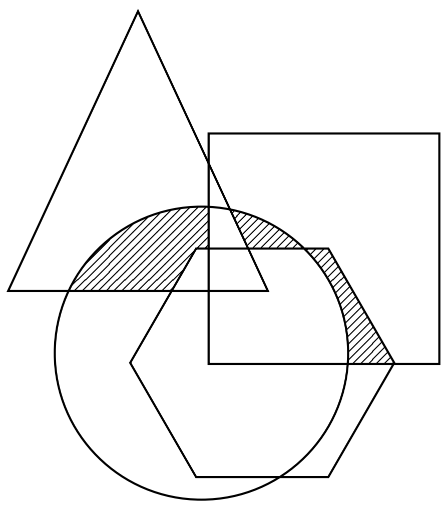
**Problem 1 part 2)**

To take a picture for fireworks at night, give at least two operations that should be applied to improve the visualization quality of the image and explain. (15 points)

**Problem 2)**

A, B, C and D are four sets. Give expressions for the set shown shaded in Figure 2 in terms of A, B, C, and D. Note that all of the shaded areas constitute of one set



**Problem 3)**

Given a 3x3 spatial filter $\frac{1}{9}\begin{bmatrix} -1 & -1 & -1 \\ -1 & -17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$ answer the following questions:

a) What type of filter is it? When applying this filter on an image, what kind of effect you will expect in the output image? Why? (Hint: is this filter an image smoothing filter, an image sharpening filter or an edge detector?) (10 points)

b) Perform the image convolution using this filter on a 3x3 image $\begin{bmatrix} 3 & 5 & 12 \\ 7 & 18 & 11 \\ 5 & 9 & 15 \end{bmatrix}$. Give the CROPPED filtering result for the output image. (Hint: you need to assume an appropriate zero mapping) (15 points)

**Problem 4)**

An image with intensities in the range $r \in [0, L-1]$ has the pdf

$$p_r(r) = \begin{cases} \dfrac{4r}{3(L-1)^2} & 0 \leq r \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

where $L-1$ is the maximum intensity value. It is desired to transform the intensity levels of this image so that they will have the specified pdf

$$p_z(z) = \begin{cases} \dfrac{2z}{(L-1)^2} & 0 \leq z \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

Assume that the intensity value is continuous and find the transformation function $z = T(r)$ in terms of $r$ and $z$ that will accomplish this. (25 points)

(Hints: The transformation function of histogram equalization is given as

$$s = T(r) = (L-1) \int_0^r p_r(w)\, dw$$

)

## Bonus Question

Perform the Fourier transform to a function as shown in Figure 3. (20 points)

(Hint: the definition of the Fourier transform of a function is $F(u) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi ut}\,dt$ )